NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

PROJECT REPORT ON

"CLASSIFICATION OF ORANGE COLORS USING MACHINE LEARNING"

AS A COMPONENT OF

Introduction to Artificial Intelligence and Machine Learning AM1502-1

Submitted by

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UNDER THE GUIDANCE OF

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1. Description of Dataset

The dataset used in this project is taken from Kaggle and contains information on different colors of oranges. It includes features such as Size (cm) ,Weight (g),Brix etc which are used to classify oranges. The target variable is the orange color. The dataset has 5 classes: **Deep Orange, Light Orange, Orange, Orange-Red, and Yellow-Orange**, with a total of 164 instances. It is used for classification using machine learning algorithms.

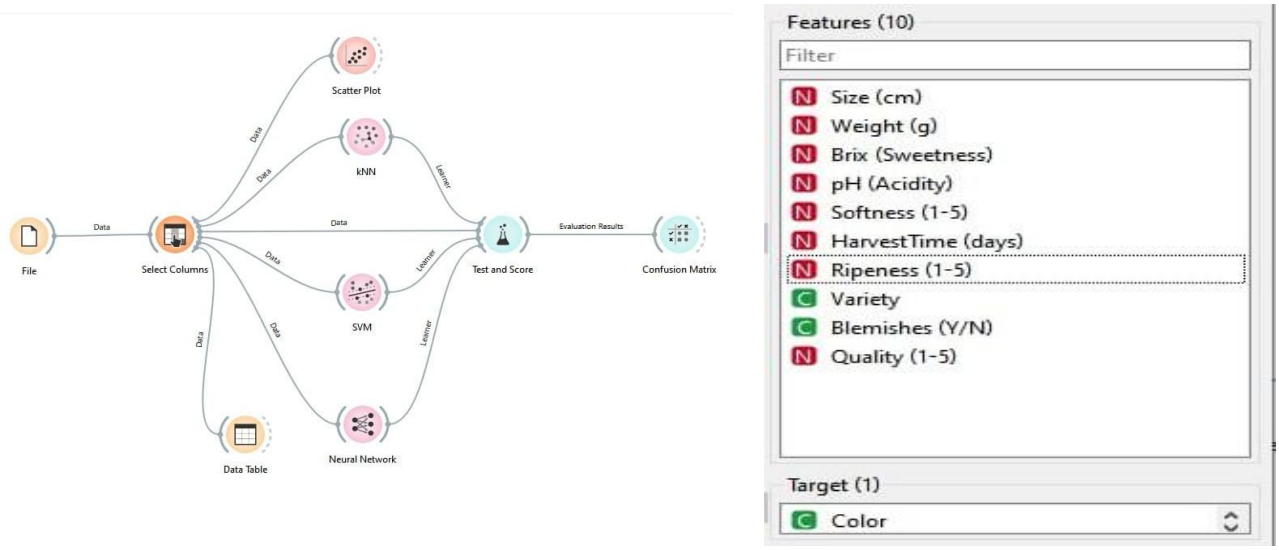


Fig 1: Workflow of Orange Tool for Classification , Features and Target

2. Layout of the Model

The workflow of the model is designed using the Orange Color as follows:

- **File Widget**: Used to load the dataset
- **Select Columns**: Used to select input features (Size, Weight, Brix, etc) and target class (orange color)
- **Scatter Plot**: Used to visualize the relationship between variables
- **KNN, SVM, Neural Network**: These are the machine learning algorithms used for classification
- **Test and Score**: Used to evaluate the performance of models
- **Confusion Matrix**: Used to analyze classification results

This layout helps in building, testing, and comparing multiple models efficiently.

3. Evaluation Scope (Algorithms Used)

K-Nearest Neighbors (KNN)

KNN is a simple algorithm that classifies data based on the nearest neighbors. It works well for smaller datasets but may give errors when classes are similar.

Support Vector Machine (SVM)

SVM is a powerful algorithm that separates data using an optimal boundary. It provides better accuracy and performs well for classification problems.

Neural Network

Neural Network is an advanced algorithm that can handle complex patterns in data. It gives good results but may require proper tuning.

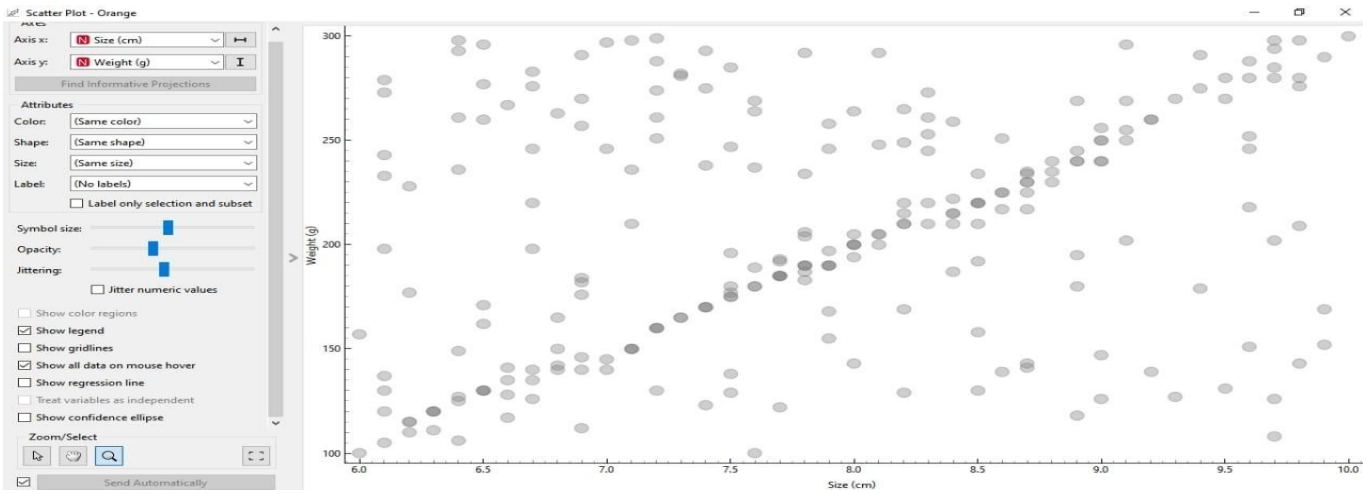
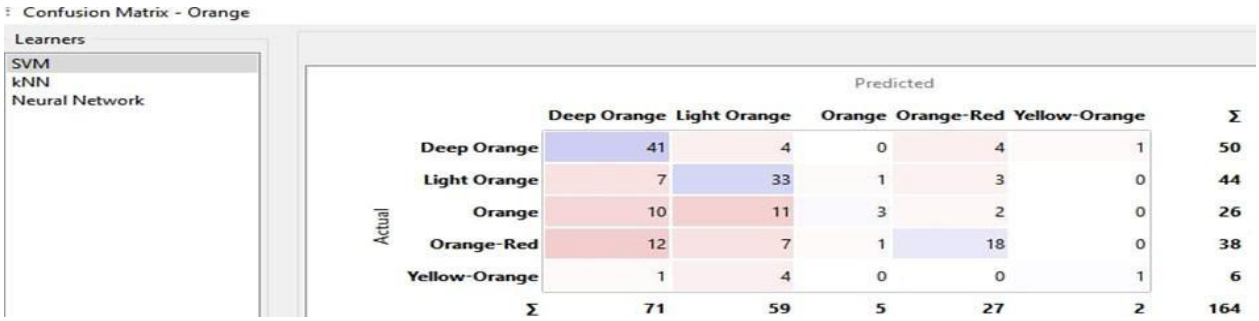


Fig 2: Scatter Plot showing Size vs Weight

4. Confusion Matrix Values

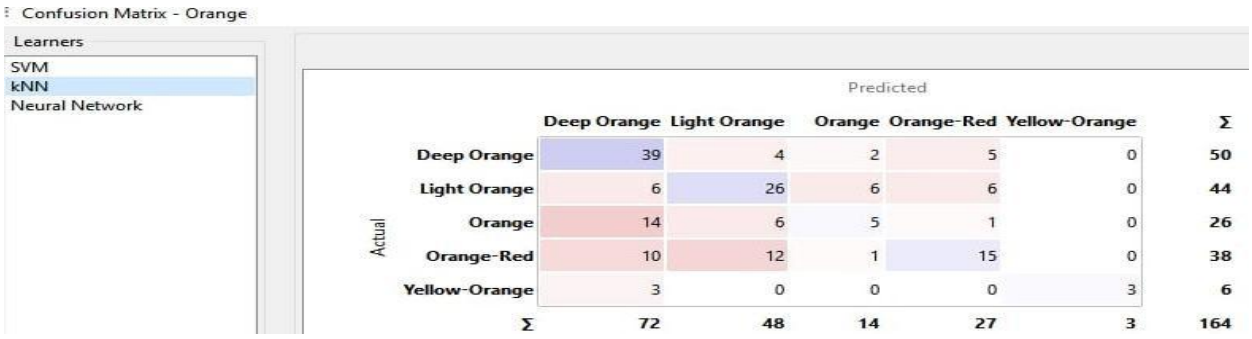
The confusion matrix is used to evaluate the performance of the models.

- The **diagonal values represent correct predictions**
- The **non-diagonal values represent misclassification**



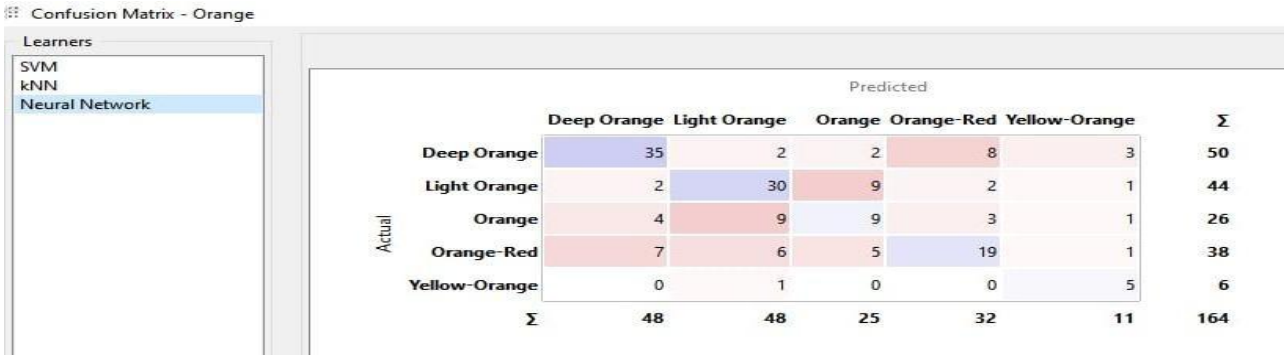
Actual \ Predicted	Deep Orange	Light Orange	Orange	Orange-Red	Yellow-Orange	Σ
Deep Orange	41	4	0	4	1	50
Light Orange	7	33	1	3	0	44
Orange	10	11	3	2	0	26
Orange-Red	12	7	1	18	0	38
Yellow-Orange	1	4	0	0	1	6
Σ	71	59	5	27	2	164

Fig 3: Confusion Matrix of SVM



Actual \ Predicted	Deep Orange	Light Orange	Orange	Orange-Red	Yellow-Orange	Σ
Deep Orange	39	4	2	5	0	50
Light Orange	6	26	6	6	0	44
Orange	14	6	5	1	0	26
Orange-Red	10	12	1	15	0	38
Yellow-Orange	3	0	0	0	3	6
Σ	72	48	14	27	3	164

Fig 4: Confusion Matrix of KNN



Actual \ Predicted	Deep Orange	Light Orange	Orange	Orange-Red	Yellow-Orange	Σ
Deep Orange	35	2	2	8	3	50
Light Orange	2	30	9	2	1	44
Orange	4	9	9	3	1	26
Orange-Red	7	6	5	19	1	38
Yellow-Orange	0	1	0	0	5	6
Σ	48	48	25	32	11	164

Fig 5: Confusion Matrix of Neural Network

5. Conclusion

In this project, orange color classification is performed using the Orange data mining tool with KNN, SVM, and Neural Network algorithms.
Among these models, SVM gives the best performance with the highest accuracy.
Scatter plots help visualize relationships between features, and the confusion matrix evaluates model performance.
Thus, machine learning techniques in Orange are effective for solving classification problems.